

Input Social Media Post (Taglish)
"Baha na po dito sa Brgy. Sto. Domingo!
Stranded mga tao, please rescue us."

Gemma-3 SentencePiece Tokenizer
Vocabulary size $V = 262,144$ subword tokens

Shared Input Embedding Layer ($V \times d = 262,144 \times 768$)
 201.3×10^6 parameters (approx. 64.4% of model total)
32-bit Floating Point Memory: ≈ 805 MB

Memory Footprint Benefit:

- **Multi-task (shared):** 805 MB (single copy)
- **Single-task baselines:**
Triplicated: $3 \times 805 \text{ MB} \approx 2,415 \text{ MB}$
- **Memory reduction:** $\approx 1,610 \text{ MB}$ (66.7%)

Shared Multilingual Encoder:
SEA-LION-ModernBERT-300M
22 Hidden Layers · Hidden Dimension $d = 768$ · 12 Attention Heads
Rotary Position Embeddings (RoPE) · FlashAttention · Unpadding
Non-embedding parameter budget: $\approx 111.2 \times 10^6$ parameters

Task-Specific Prediction Heads (Identical in MTL and Baseline STL)

NER Token Head
(Token Level)
Dropout $\rightarrow$ Linear
 $768 \rightarrow |\mathcal{Y}_{\text{NER}}|$
BIO tags: LOC, INFRA

Intent Head
(Sequence Level)
Mean Pooling $\rightarrow$ Linear
 $768 \rightarrow 8$ classes
 $|\mathcal{Y}_{\text{Intent}}| = 8$

Urgency Head
(Sequence Level)
Mean Pooling $\rightarrow$ Linear
 $768 \rightarrow 3$ classes
 $|\mathcal{Y}_{\text{Urgency}}| = 3$

Multi-Task Balancing and Loss Aggregation

$\mathcal{L}_{\text{total}} = w_1 \mathcal{L}_{\text{NER}} + w_2 \mathcal{L}_{\text{Intent}} + w_3 \mathcal{L}_{\text{Urgency}}$
Evaluated under: Uniform ($w_t = 1$), Static
Grid Search (w_t^*), and PCGrad Gradient Surgery